

ICT Course: Introduction to Cryptography

Nguyen Minh Huong

ICT Department, USTH

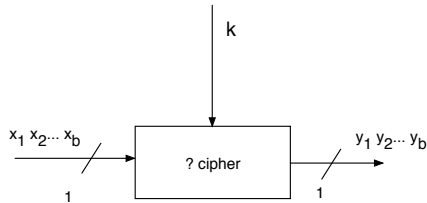
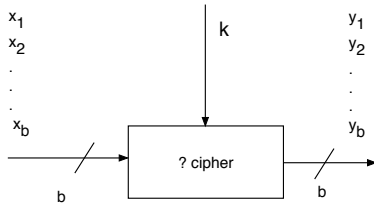
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Session 3: Introduction to Cryptography - Symmetric Cryptography

- 1 Symmetric ciphers
 - Stream ciphers
 - A5/1 cipher
 - RC4 - Keystream generation

Symmetric ciphers

Stream cipher vs. Block cipher



Stream Cipher-How it works

- A key K of n bits is stretched into a long keystream S

$$\text{StreamCipher}(K) = S$$

Stream cipher Encryption and Decryption

$x_i, y_i, s_i \in \{0, 1\}$: individual bits of plaintext, ciphertext and keystream

- Encryption: $y_i = x_i + s_i \pmod{2}$
- Decryption: $x_i = y_i + s_i \pmod{2}$

Stream ciphers

- Modulo 2 addition is equivalent to XOR operation
- Plaintext P , Ciphertext C

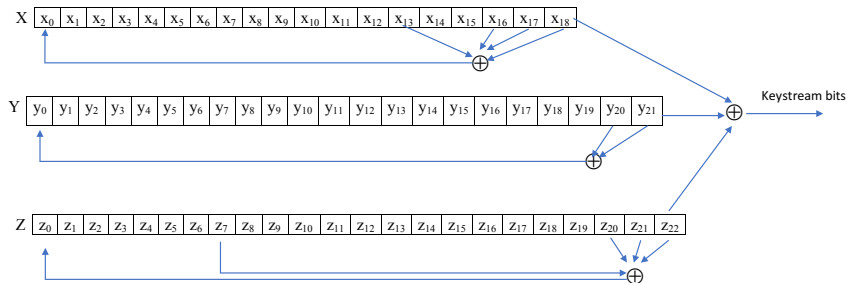
$$C = P \oplus S$$

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A5/1 cipher

- designed for hardware, used in GSM
- Key of 64 bits initially fills in 3 Linear Feedback Shift Registers (LFSRs):
 - X 19 bits
 - Y 22 bits
 - Z 23 bits
- How to obtain every single keystream bit?

A5/1 - Keystream generator



A5/1 - Keystream generator

- LFSR steps:
 - X steps then:

$$x_i = x_{i-1}$$

$$x_0 = x_{13} \oplus x_{16} \oplus x_{17} \oplus x_{18}$$

- Y steps then: ?
 - Z steps then: ?
- A single keystream bit: $s = ?$

A5/1 - Keystream generation

When they step?

- Majority vote function: $m = \text{maj}(x, y, z)$, $m = 0$ when majority of (x, y, z) is 0, otherwise $m = 1$
- In A5/1, $m = \text{maj}(x_8, y_{10}, z_{10})$
 - if $x_8 = m$, then X steps
 - if $y_{10} = m$, then Y steps
 - if $z_{10} = m$, then Z steps

RC4 algorithm

- Designed for software, e.g, SSL, WEP...
- Key: N bytes, $1 \leq N \leq 256$
- Lookup table: a 256-byte array S
 - Initialized from the key to the identify permutation: Key-scheduling Algorithm (KSA)
 - can be one of all 256 possible permutations of 256 bytes
- From the lookup table, a keystream byte is generated at each step: Pseudo-random generation algorithm (PRGA)

RC4 - Keystream generation

- KSA pseudo code:

```
1: for  $i = 1$  to 255 do
2:    $S[i] = i$ 
    $K[i] = \text{key}[i \bmod N]$ 
3: end for
    $j=0$ 
4: for  $i = 0$  to 255 do
5:    $j = (j + S[i] + K[i]) \bmod 256$ 
   Swap( $S[i]$ ,  $S[j]$ )
6: end for
    $i=j=0$ 
```

RC4 - Keystream generation

- PRGA pseudo code:

$$i = (i + 1) \bmod 256$$

$$j = (j + S[i]) \bmod 256$$

$$\text{Swap}(S[i], S[j])$$

$$t = (S[i] + S[j]) \bmod 256$$

$$\text{KeystreamByte} = S[t]$$